

LIMITS OF MATRIX APPROXIMATION OF GROUPS IN OPERATOR NORM

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We show that not every countable group is MF, where MF means admitting finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. The main tool is a one-sided compression criterion that places normal Kazhdan subgroups in the MF radical. Applied to the binary Leavitt group, it shows that the simple property-(T) group $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$ has trivial image under every homomorphism to an MF group; consequently its reduced group C^* -algebra is a separable stably finite C^* -algebra that is not MF. A second application gives a finitely presented torsion-free group with property (T) that has trivial image under every homomorphism to an MF group, and every nontrivial quotient of it has the same property.

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1. INTRODUCTION

Not every countable group is MF. Our counterexample is

$$H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2)).$$

The group H is countable, nontrivial, simple, and has property (T), but every homomorphism from H to an MF group is trivial. In particular, H is not MF. Its reduced group C^* -algebra $C_r^*(H)$ is separable and stably finite, but is not MF.

More precisely, a countable group G is *MF* if there are positive integers d_n and maps $V_n: G \rightarrow \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE].

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Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \rightarrow 0\}$$

is the c_0 -direct sum.

A separable C^* -algebra is called MF if it embeds as a C^* -subalgebra of a norm matrix corona. For every countable group G , the reduced group algebra $C_r^*(G)$ is separable and has a faithful canonical trace, hence is stably finite. Suppose that $e: C_r^*(G) \rightarrow \mathcal{Q}_{\mathbf{d}}$ is an MF embedding and put $p = e(1)$. Then $g \mapsto e(\lambda_g) + (1 - p)$ is an injective homomorphism from G into $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$, so G is MF. Therefore a non-MF group automatically gives a separable stably finite reduced group C^* -algebra that is not MF.

For a countable group G , define its MF radical by

$$\text{Rad}_{\text{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

We call a homomorphism $G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. Proposition 2.1 shows that G is MF precisely when $\text{Rad}_{\text{MF}}(G)$ is trivial. If $\text{Rad}_{\text{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

For a subgroup $L \leq G$, put

$$\text{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression-centralizer defect* by

$$(1) \quad \mathfrak{D}_G(L) = \langle\langle [ucu^{-1}, \ell] : u \in \text{Comp}_G(L), c \in C_G(L), \ell \in L \rangle\rangle_G.$$

Theorem A (one-sided compression criterion). *Let G be countable and let $L \leq G$ have property (T). If $K \trianglelefteq G$ has property (T) and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF. If G has property (T) and $\mathfrak{D}_G(L) = G$, then $\text{Rad}_{\text{MF}}(G) = G$.

We sketch the argument. Corollary 2.6 gives $\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G)$. That is, every operator norm asymptotic representation of G is asymptotically trivial on $\mathfrak{D}_G(L)$ in normalized Hilbert–Schmidt norm. If a corona homomorphism were nontrivial on K , the Kazhdan projection of K would cut out a G -invariant corner on which K has no nonzero fixed vectors (Lemma 2.7); by the Kazhdan inequality some $s_0 \in K$ would then stay bounded away from the identity in normalized Hilbert–Schmidt norm along a subsequence, contradicting $s_0 \in \mathfrak{D}_G(L)$.

Theorem 2.2 gives a separate finite-dimensional vanishing result: every finite-dimensional linear representation of G over any field kills $\mathfrak{D}_G(L)$.

We next apply Theorem A to the group H .

Let $L_{\mathbb{F}_2}(1, 2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$ -algebra generated by s_0, s_1, t_0, t_1 subject to

$$(2) \quad t_i s_j = \delta_{ij}, \quad s_0 t_0 + s_1 t_1 = 1.$$

For a unital ring R , write $e_{ij}(a) = 1 + aE_{ij}$ and let $\text{EL}_n(R)$ be the subgroup of $\text{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group). *Put $R = L_{\mathbb{F}_2}(1, 2)$ and $H = \text{EL}_{12}(R)$. Then H is countable, nontrivial, simple, has property (T), and*

$$\text{Rad}_{\text{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF. Its reduced group C^ -algebra $C_r^*(H)$ is separable and stably finite, but is not MF.*

Inside H , an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the subgroup $L \cong \text{EL}_3(R)$ in the upper left. The element $c = e_{34}(1)$ centralizes L , and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The relation $t_1(s_1 t_1)s_1 = 1$ shows that $d \neq 1$. Since H is simple, d normally generates H . Since $d \in \mathfrak{D}_H(L)$ and $\mathfrak{D}_H(L)$ is normal, it follows that $\mathfrak{D}_H(L) = H$. Therefore Theorem A gives $\text{Rad}_{\text{MF}}(H) = H$.

The second application of Theorem A is to a group without torsion.

Theorem C (a torsion-free group with full MF radical). *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group Q with property (T) and*

$$\text{Rad}_{\text{MF}}(Q) = Q.$$

Every nontrivial quotient of Q also equals its own MF radical; in particular, no nontrivial quotient of Q is MF.

The group Q is a small-cancellation quotient, in the sense of Hull [Hu], of the torsion-free property-(T) group of Fournier-Facio [FF, §2], chosen so that the compression defect normally generates the quotient (Section 4). Thus the obstruction of Theorem A depends neither on torsion, nor on simplicity, nor on the Leavitt algebra.

In a companion paper [Sa] we use H , together with an MF permanence theorem for HNN extensions, to show that recognizing MF groups from finite presentations is Π_2^0 -complete. The companion notes [SaN] construct an explicit finitely presented group E from the ascending HNN extension of $\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$ by the doubling endomorphism of $\mathbb{Z}^3$, and prove that E is sofic and not MF; so among finitely presented groups, MF is strictly stronger than sofic and than hyperlinear.

Relation to prior work. Blackadar and Kirchberg introduced the notions of MF and NF algebras in a 1994 Oberwolfach abstract [BK94]. They wrote that the classes of NF, strong NF, and separable nuclear stably finite C^* -algebras might coincide. Their 1997 paper developed these notions and proved that a separable C^* -algebra is NF if and only if it is nuclear and MF [BK]. It also asked whether every separable nuclear stably finite C^* -algebra is quasidiagonal. This was a question about nuclear C^* -algebras, not about arbitrary groups.

The group property used here was introduced by Carrión–Dadarlat–Eckhardt [CDE]. If every separable stably finite C^* -algebra were MF, then every countable group would be MF because its reduced group C^* -algebra is separable and stably finite. Later papers on approximation of groups relate the resulting operator norm problem to Kirchberg’s question [LO, Th]. The negative solution of the Connes embedding problem [JNVWY] yields separable stably finite C^* -algebras that are not MF, as Goldbring and Hart observe [GH, Proposition 6.1 and Remark 6.2]. Theorem B gives such an example among reduced group C^* -algebras. Korchagin subsequently developed permanence and local approximation results for MF groups [Kor]. Shulman proves that $A *_C A$ is MF whenever A is a separable MF C^* -algebra and $C \subseteq A$ is a C^* -subalgebra [Sh, Theorem 10]. She also proves that the full group C^* -algebras of amalgamated free products of amenable groups and of certain semidirect products are MF [Sh]. Nonapproximability was known for the unnormalized Schatten p -norms with $1 < p < \infty$ [DGLT, LO]. Bachner–Dogon–Lubotzky later proved nonapproximability for $p = 1$ [BDL]. The operator norm case $p = \infty$ remained open. They also identify stability from operator norm to normalized Hilbert–Schmidt norm as a possible route to a non-MF group [BDL, Proposition 1.5]; that is the route taken here.

Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. The compression used here comes from their defining relations. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser classified the subgroups of GL_n over an exchange ring that are normalized by EL_n [Pre]. His sandwich theorem, together with the simplicity of the binary Leavitt algebra and the equality $Z(R) = \mathbb{F}_2$, implies that $EL_{12}(R)$ is simple (Proposition 3.3). Ershov–Jaikin–Zapirain give property (T) for $EL_n(R)$ whenever $n \geq 3$ and R is a finitely generated unital associative ring [EJZ, Theorem 1.1]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

2. ONE-SIDED COMPRESSION AND THE MF RADICAL

This section proves Theorem A. Let $uLu^{-1} \leq L$. In a finite-dimensional representation ρ , conjugation by $\rho(u)$ maps the commutant of $\rho(L)$ into itself; the map is injective, hence surjective, so the commutant is invariant (Theorem 2.2). In a norm matrix corona the same conclusion is obtained

for the Kazhdan projection of a property-(T) subgroup: conjugation moves the projection below itself, and stable finiteness of the corona forces equality (Section 2.3). A normal property-(T) subgroup contained in the compression defect then lies in the MF radical (Theorem 2.8).

2.1. The MF radical. For $N \trianglelefteq G$ define its *MF kernel closure* by

$$\text{cl}_{\text{MF}}^G(N) = \bigcap \{\ker f : N \leq \ker f, f: G \rightarrow M, M \text{ MF}\}.$$

The image of a corona homomorphism from G is countable and is itself MF; conversely, every MF target embeds in a norm matrix corona. Thus $\text{Rad}_{\text{MF}}(G) = \text{cl}_{\text{MF}}^G(1)$, and G/N is MF precisely when $\text{cl}_{\text{MF}}^G(N) = N$. This is a closure operation on the normal subgroups of one fixed group; it is unrelated to topological closure of classes of marked groups.

Proposition 2.1 (basic properties of the MF radical). *For every countable group G , the subgroup $\text{Rad}_{\text{MF}}(G)$ is fully invariant, the quotient $G/\text{Rad}_{\text{MF}}(G)$ is MF, and*

$$G/N \text{ is MF} \iff \text{cl}_{\text{MF}}^G(N) = N \quad (N \trianglelefteq G).$$

In particular, G is MF if and only if its MF radical is trivial.

Proof. An intersection of kernels is normal. If α is an endomorphism of G , $x \in \text{Rad}_{\text{MF}}(G)$, and π is a corona homomorphism from G , then $\pi \circ \alpha$ kills x . Hence $\pi(\alpha(x)) = 1$ for every π , which proves full invariance.

Put $R = \text{Rad}_{\text{MF}}(G)$. If G/R is trivial, there is nothing to prove. If $x \in G/R$ is nontrivial and $g \in G$ maps to x , then $g \notin R$. Some homomorphism from G to an MF group therefore separates g from the identity. It kills R and descends to a homomorphism from G/R that separates x . Thus G/R is residually MF, and hence MF by Korchagin [Kor, Proposition 6].

For $N \trianglelefteq G$, homomorphisms from G/N to MF groups correspond to homomorphisms from G to MF groups that kill N . It follows directly that

$$\text{Rad}_{\text{MF}}(G/N) = \text{cl}_{\text{MF}}^G(N)/N.$$

Since a countable group is MF exactly when its MF radical is trivial, this identity gives the asserted closure criterion. $\square$

2.2. Finite dimension.

Theorem 2.2 (finite-dimensional commutant rigidity). *Let G be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If k is a field, V is a finite-dimensional k -vector space, and $\rho: G \rightarrow \text{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\text{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \quad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uh u^{-1} \in L$, then

$$\begin{aligned} \rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u) \\ &= \rho(u)^{-1}x\rho(h')\rho(u) \\ &= \rho(u)^{-1}x\rho(u)\rho(h). \end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. Therefore $\rho([ucu^{-1}, \ell]) = 1$ for every $\ell \in L$. $\square$

The finite-dimensionality hypothesis cannot be dropped: an injective linear self-map of an infinite-dimensional commutant need not be surjective.

2.3. Kazhdan transport in normalized Hilbert–Schmidt norm. For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \text{tr}_d(a^*a)^{1/2}, \quad \text{tr}_d = \frac{1}{d} \text{Tr}.$$

An *operator norm asymptotic representation* of a countable group G is a sequence of maps $V_n: G \rightarrow \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \rightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G . Define

$$(3) \quad R_{\infty \rightarrow 2}(G) = \bigcap_V K_2(V),$$

where V ranges over all operator norm asymptotic representations of G .

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \left| \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \rightarrow 0 \quad (\ell \in L) \end{array} \right. \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatwise conjugation by $V_n(g)$ defines a bijection, denoted $\text{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 2.3. *For every sequence of positive integers (m_n) , the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v . Then $x_n^*x_n \rightarrow 1$ in norm. For all sufficiently large n , the matrix x_n is invertible, and the unitary $x_n(x_n^*x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. If A is a stably finite unital C^* -algebra, equivalent projections $p \leq q$ satisfy $p = q$. Indeed, if $w^*w = q$ and $ww^* = p$, then $w \in qAq$ is an isometry in the corner. The corner is finite because $w + (1 - q)$ is an isometry in A . Consequently $ww^* = q$, and hence $p = q$. $\square$

Lemma 2.4 (one-sided order for the Kazhdan projection). *Let L have property (T), let B be a unital C^* -algebra, and let $\pi: L \rightarrow \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C_{\max}^*(L) \rightarrow B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \leq P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\text{Fix } \pi(L)$. If ξ is L -fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^*\text{Fix } \pi(L) \subseteq \text{Fix } \pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B . $\square$

Theorem 2.5 (one-sided Kazhdan transport). *Let $L \leq G$ have property (T), and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator norm asymptotic representation of G . Then*

$$\text{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V, L)$ and let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\text{Ad}(A) - \text{Ad}(B)\| \leq 2\|A - B\|,$$

so the conjugation actions are multiplicative modulo c_0 because the multiplicative defects of (V_n) tend to zero in operator norm. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\tilde{\sigma}(g)$ be the class of the coordinate operators $\text{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2 . By the preceding estimate, $g \mapsto \tilde{\sigma}(g)$ is a homomorphism $G \rightarrow \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$ -homomorphism $C_{\max}^*(L) \rightarrow \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\text{Fix } \sigma(L)$. The projection identity below is proved inside $\mathcal{B}$; the representation of $\mathcal{B}$ on $\mathcal{K}_\omega$ is used only afterwards. Since $uLu^{-1} \leq L$, the unitary $U = \tilde{\sigma}(u)$ satisfies the hypothesis of Lemma 2.4, and hence $U^*PU \leq P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 2.3 therefore gives $U^*PU = P$, so U commutes with P . Hence both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus $\text{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$, which proves the equality. $\square$

Corollary 2.6. *Let $L \leq G$ have property (T), let $uLu^{-1} \leq L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G) \quad (\ell \in L).$$

Proof. Let (V_n) be an operator norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V, L)$. By Theorem 2.5, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \longrightarrow 0.$$

Therefore

$$\|V_n([ucu^{-1}, \ell]) - 1\|_2 \longrightarrow 0 \quad (\ell \in L).$$

Intersecting over (V_n) proves the claim. $\square$

2.4. Normal Kazhdan subgroups and the MF radical. Let $K \trianglelefteq G$ have property (T), and let p be the image of its Kazhdan projection under a corona homomorphism of G . Normality of K implies that p commutes with the image of G . Hence $q = 1 - p$ also commutes with that image, and in every nondegenerate representation of the corner $q\mathcal{Q}_{\mathbf{d}}q$ the induced representation of K has no nonzero fixed vectors.

Lemma 2.7 (central corona corners). *Let $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \text{rank}(q_n)$, unitary identifications $J_n: \mathbb{C}^{r_n} \rightarrow q_n\mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n: G \longrightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$. In the corona over the retained coordinates, the class of $(J_n W_n(g) J_n^)$ is the coordinate restriction of $q\rho(g)$.*

Proof. Lift q to projections q_n , and lift each $\rho(g)$ to unitaries $U_n(g)$, choosing $U_n(1) = I_{d_n}$. The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \rightarrow 0 \quad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n \mathbb{C}^{d_n}$, has both unitarity defects converging to zero. For each fixed g , let $\widetilde{W}_n(g)$ be its polar correction whenever both defects are at most $1/2$, and set $\widetilde{W}_n(g) = q_n$ at the finitely many earlier stages. Then $\widetilde{W}_n(g)$ is unitary in the corner and satisfies

$$\|\widetilde{W}_n(g) - q_n U_n(g) q_n\| \rightarrow 0.$$

For $g = 1$, the compression is already the identity q_n of the corner, so take $\widetilde{W}_n(1) = q_n$. Since $q \neq 0$, infinitely many q_n are nonzero. Retain and relabel those coordinates. Put $r_n = \text{rank}(q_n)$, choose a unitary $J_n: \mathbb{C}^{r_n} \rightarrow q_n \mathbb{C}^{d_n}$, and define

$$W_n(g) = J_n^* \widetilde{W}_n(g) J_n \in \mathcal{U}(r_n).$$

Conjugation by J_n preserves operator norms and multiplicative defects. Thus (W_n) is an operator norm asymptotic representation with $W_n(1) = I_{r_n}$. The displayed estimate identifies the class of $(J_n W_n(g) J_n^*)$ with the coordinate restriction of $q\rho(g)$. $\square$

Theorem 2.8 (normal Kazhdan radical theorem). *Let G be countable and let $K \trianglelefteq G$ have property (T) with $K \leq R_{\infty \rightarrow 2}(G)$. Then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K . Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ . The new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. It is contained in $R_{\infty \rightarrow 2}(\Theta(G))$, because an operator norm asymptotic representation of $\Theta(G)$, precomposed with the quotient map, is one of the original group.

Let $p_K \in C_{\max}^*(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Under any faithful representation of the corona, p is the projection onto the K -fixed vectors. Normality of K gives

$$\Theta(g) p \Theta(g)^* = p \quad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges $\Theta(g) \text{Fix } \Theta(K)$ and $\text{Fix } \Theta(K)$; these agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection q is nonzero: if $p = 1$, then every vector is fixed by K , so every element of K has trivial corona image. Lemma 2.7 gives positive integers r_n and an operator norm asymptotic representation $W_n: G \rightarrow \mathcal{U}(r_n)$. Let $\widehat{\Theta}: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{r}})$ be its corona homomorphism. Under the coordinatewise corner identifications, $\widehat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$.

Choose a finite Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. Represent $\mathcal{Q}_r$ faithfully. The Kazhdan projection of K under $\hat{\Theta}$ is zero, so $\hat{\Theta}|_K$ has no nonzero fixed vectors, and the Kazhdan inequality gives, for the positive element

$$b = \frac{1}{|S|} \sum_{s \in S} (\hat{\Theta}(s) - 1)^* (\hat{\Theta}(s) - 1)$$

the inequality

$$b \geq \frac{\kappa^2}{|S|} 1.$$

The coordinate elements

$$b_n = \frac{1}{|S|} \sum_{s \in S} (W_n(s) - I_{r_n})^* (W_n(s) - I_{r_n})$$

represent b . Taking the normalized trace on $M_{r_n}(\mathbb{C})$ yields

$$\frac{1}{|S|} \sum_{s \in S} \|W_n(s) - I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)I_{r_n}$ converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed $s_0 \in S$ therefore stays a positive Hilbert–Schmidt distance from I_{r_n} . This contradicts $s_0 \in K \leq R_{\infty \rightarrow 2}(G)$, established after passing to $\Theta(G)$. Thus every corona homomorphism kills K , as required. $\square$

Proof of Theorem A. For every $u \in \text{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 2.6 gives

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G).$$

Normality of $R_{\infty \rightarrow 2}(G)$ therefore gives

$$\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G).$$

Since $K \leq \mathfrak{D}_G(L)$, Theorem 2.8 now gives $K \leq \text{Rad}_{\text{MF}}(G)$. Taking a nontrivial K gives the non-MF conclusion. Taking $K = G = \mathfrak{D}_G(L)$ gives $\text{Rad}_{\text{MF}}(G) = G$.

A finite-dimensional linear representation over any field kills every generator of $\mathfrak{D}_G(L)$ by Theorem 2.2; its kernel is normal, so it contains $\mathfrak{D}_G(L)$. $\square$

If $\rho: G \rightarrow \text{GL}(V)$ is a homomorphism and V is finite-dimensional, conjugation by $\rho(u)^{-1}$ maps the commutant of $\rho(L)$ injectively into itself, hence bijectively. Thus ρ kills $\mathfrak{D}_G(L)$. Consequently, $\mathfrak{D}_G(L) = 1$ if G has a faithful finite-dimensional linear representation. The same conclusion holds if G is residually finite, because every nonidentity element then survives in a finite permutation representation. If G is amenable, its property-(T) subgroup L is finite [BHV]; then $uLu^{-1} = L$, so ucu^{-1} centralizes L for every $u \in \text{Comp}_G(L)$, and again $\mathfrak{D}_G(L) = 1$.

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt

approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 2.5.

3. THE BINARY LEAVITT GROUP

This section proves Theorem B. The relation $s_0t_0 + s_1t_1 = 1$ of the binary Leavitt algebra gives an element $\tau \in H$ with $\tau L \tau^{-1} \leq L$ for a copy L of $\text{EL}_3(R)$, and a commutator $d \neq 1$ in $\mathfrak{D}_H(L)$; since H is simple, $\mathfrak{D}_H(L) = H$, and Theorem A applies.

3.1. The self-compression. Throughout this section $R = L_{\mathbb{F}_2}(1, 2)$ and

$$p = s_0t_0, \quad q = s_1t_1.$$

The relations (2) give

$$(4) \quad p + q = 1, \quad t_1qs_1 = 1.$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for 12×12 matrices. We identify $\text{GL}_3(R)$ with the subgroup of $\text{GL}_{12}(R)$ consisting of the matrices $\text{diag}(A, I_9)$. Define a map $\Psi: M_3(R) \rightarrow M_3(R)$ by

$$(5) \quad \Psi(A) = qI_3 + s_0At_0.$$

Here s_0At_0 denotes the matrix $(s_0a_{ij}t_0)_{ij}$. The relations $q^2 = q$, $qs_0 = t_0q = 0$, and $t_0s_0 = 1$, all consequences of (2), show that Ψ is multiplicative. Moreover,

$$\Psi(I_3) = qI_3 + s_0I_3t_0 = qI_3 + pI_3 = (p + q)I_3 = I_3,$$

so Ψ preserves the identity. Finally, $t_0\Psi(A)s_0 = A$, so Ψ is injective. Hence Ψ restricts to an injective group homomorphism $\text{GL}_3(R) \rightarrow \text{GL}_3(R)$.

To realize Ψ by conjugation on the embedded copy of $\text{GL}_3(R)$, define the following 6×6 block matrices:

$$X = \begin{pmatrix} s_0I_3 & s_1t_0I_3 \\ 0 & t_1I_3 \end{pmatrix}, \quad Y = \begin{pmatrix} t_0I_3 & 0 \\ s_0t_1I_3 & s_1I_3 \end{pmatrix}.$$

A block multiplication using (2) gives $XY = YX = I_6$, so $Y = X^{-1}$. Put

$$(6) \quad \tau = \text{diag}(X, Y) \in \text{GL}_{12}(R).$$

A direct block multiplication gives

$$(7) \quad \tau \text{diag}(A, I_9) \tau^{-1} = \text{diag}(\Psi(A), I_9).$$

Lemma 3.1. *The matrix τ defined in (6) belongs to $\text{EL}_{12}(R)$.*

Proof. Put $I = I_6$. The Whitehead factorization gives

$$(8) \quad \text{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}.$$

If $B = (b_{rs})_{r,s=0}^5 \in M_6(R)$, then

$$\begin{pmatrix} I & B \\ 0 & I \end{pmatrix} = \prod_{r,s=0}^5 e_{r,6+s}(b_{rs}), \quad \begin{pmatrix} I & 0 \\ B & I \end{pmatrix} = \prod_{r,s=0}^5 e_{6+r,s}(b_{rs}).$$

In either product, distinct off-diagonal matrix units have product zero, so no cross-terms occur. Every factor on the right-hand side of (8) therefore belongs to $\text{EL}_{12}(R)$. Since $Y = X^{-1}$, its left-hand side is τ , and the result follows. $\square$

Set $H = \text{EL}_{12}(R)$, and let $L = \text{EL}_3(R) \leq H$ denote the image of the embedding $A \mapsto \text{diag}(A, I_9)$.

Proposition 3.2. *The groups H and L have property (T), and $\tau \in H$ satisfies*

$$(9) \quad \tau L \tau^{-1} \leq L.$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring. The theorem of Ershov and Jaikin-Zapirain therefore gives property (T) for $\text{EL}_{12}(R)$ and $\text{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 3.1 gives $\tau \in H$.

For $i \neq j$ and $a \in R$, equations (7) and (5) give

$$\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0 a t_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of L . These matrices generate L , so (9) follows. $\square$

3.2. Simplicity and the full MF radical.

Proposition 3.3. *The group $H = \text{EL}_{12}(R)$ is nontrivial and simple.*

Proof. The ring R is purely infinite simple [AA], and therefore is an exchange ring [Ara]. Let $N \trianglelefteq H$. Since $H = \text{EL}_{12}(R)$, the subgroup $N \leq \text{GL}_{12}(R)$ is normalized by $\text{EL}_{12}(R)$. Preusser's sandwich theorem [Pre, Theorem 3] provides an ideal $I \trianglelefteq R$ such that

$$\text{EL}_{12}(R, I) \leq N \leq C_{12}(R, I).$$

Simplicity of R gives $I = 0$ or $I = R$. If $I = R$, then $\text{EL}_{12}(R, I) = H$, so $N = H$.

Suppose that $I = 0$. Then $N \leq C_{12}(R, 0) = Z(\text{GL}_{12}(R))$. A matrix that commutes with every $e_{ij}(1)$ is a scalar matrix λI_{12} . Commutation with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a \lambda$, so $\lambda \in Z(R)^\times$. Since $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], we have $\lambda = 1$ and $N = 1$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 3.4. *Let*

$$c = e_{34}(1), \quad \ell = e_{12}(1), \quad d = [\tau c \tau^{-1}, \ell]$$

in H . Then

$$c \in C_H(L), \quad d = e_{02}(q) \neq 1, \quad \langle\langle d \rangle\rangle_H = H.$$

Proof. For $i, j \in \{0, 1, 2\}$ with $i \neq j$ and $a \in R$, we have $j \neq 3$ and $i \neq 4$. Hence the Steinberg relations give

$$[e_{ij}(a), e_{34}(1)] = 1.$$

These matrices generate L , so $c = e_{34}(1)$ belongs to $C_H(L)$.

Direct multiplication of the matrices in (6) gives

$$(10) \quad \tau c \tau^{-1} = e_{01}(q) e_{34}(1).$$

The second factor commutes with $\ell = e_{12}(1)$. The identity $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ therefore gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$. Its normal closure is a nontrivial normal subgroup of H , and hence is all of H by Proposition 3.3. $\square$

Proof of Theorem B. The ring R is a finitely generated $\mathbb{F}_2$ -algebra, so H is countable, and H has property (T) by the Ershov–Jaikin–Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (9), the centrality of c relative to L , and Proposition 3.4 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 3.4 gives $\langle\langle d \rangle\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\text{Rad}_{\text{MF}}(H) = H$. Proposition 3.3 shows that H is nontrivial and simple.

If M is MF and $f: H \rightarrow M$ is a homomorphism, compose f with a faithful corona embedding of M . Since $\text{Rad}_{\text{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion.

Finally, $C_r^*(H)$ is separable because H is countable. Its canonical trace is faithful, so it is stably finite. If an MF embedding $e: C_r^*(H) \rightarrow \mathcal{Q}_{\mathbf{d}}$ existed and $p = e(1)$, then $h \mapsto e(\lambda_h) + (1 - p)$ would embed H in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$. This contradicts the fact that H is not MF. $\square$

4. A TORSION-FREE GROUP WITH FULL MF RADICAL

A group is acylindrically hyperbolic if it admits a non-elementary acylindrical action on a hyperbolic space [Os]. For such a group G , Hull [Hu, Theorem 3.12] provides a generating set A such that the Cayley graph $\Gamma(G, A)$ is hyperbolic and the action of G on it is acylindrical and non-elementary; a subgroup $N \leq G$ is *suitable* with respect to A if N acts non-elementarily on $\Gamma(G, A)$ and normalizes no nontrivial finite subgroup of G [Hu, Definition 1.4]. We use Hull’s small cancellation theorem [Hu, Theorem 7.1] in the following form.

Theorem 4.1 (Hull’s small cancellation theorem). *Let G be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to A , and let $t_1, \dots, t_m \in G$ and a finite subset $F \subseteq G$ be given. Then there is a surjective homomorphism $q: G \rightarrow Q$ such that Q is acylindrically hyperbolic, $q|_F$ is injective, $q(t_i) \in q(N)$ for all i , and every element of finite order in Q is the image of an*

element of the same order in G . The kernel of q is the normal closure of m elements.

The last sentence is read off from Hull's proof: the case $m = 1$ adds the single relator $W = t_1^{-1}h_1^{m_1}h_2^{l_1} \cdots h_1^{m_n}h_2^{l_n}$ for suitable $h_1, h_2 \in N$, and the general case is an induction on m .

Lemma 4.2 (saturation). *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $F \subseteq G$ be finite. Then there is a surjective homomorphism $q: G \rightarrow Q$ such that Q is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $q|_F$ is injective, and $q(N) = Q$.*

Proof. Fix A as above. Since G is torsion-free and $N \neq 1$, N is infinite, so $|N \cap gNg^{-1}| = \infty$ for every $g \in G$, and by Osin [Os, Lemma 7.1] N acts non-elementarily on $\Gamma(G, A)$. The only finite subgroup of G is trivial, so N is suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], N contains two elements h_1, h_2 such that $N_0 = \langle h_1, h_2 \rangle$ is again suitable. Apply Theorem 4.1 to N_0 with $t_1, \dots, t_m$ a finite generating set of G and with the set F . Then

$$Q = \langle q(t_1), \dots, q(t_m) \rangle \leq q(N_0) \leq q(N) \leq Q,$$

so $Q = q(N_0) = \langle q(h_1), q(h_2) \rangle$ and $q(N) = Q$. The group Q is torsion-free because G is, and finitely presented because it is G with m relators added. $\square$

We now recall the construction of Fournier-Facio [FF, §2]; the references for the ingredients are given there. Let U be a universal finitely presented torsion-free group, one containing a copy of every finitely presented torsion-free group; let S be a finitely presented infinite simple torsion-free group; and let H_0 be a torsion-free hyperbolic group with property (T). Small cancellation over the relatively hyperbolic pair $(U * H_0, U)$ embeds U in a finitely presented torsion-free quotient P of H_0 , which therefore has property (T). By universality, P contains a subgroup $P_1 \times P_2 \times S$ with $P_i \cong P$. Let E be the double HNN extension of P with stable letters u_1, u_2 and $u_i P u_i^{-1} = P_i$; it is finitely presented, torsion-free, and acylindrically hyperbolic. Hull's common quotient theorem [Hu, Corollary 7.4] applied to E and H_0 gives a surjection $\pi: E \rightarrow G_0$ onto a finitely presented torsion-free group with property (T), injective on a prescribed finite subset of E ; since E is finitely generated, the proof of that corollary produces G_0 acylindrically hyperbolic. We run the construction with the prescribed finite subset containing an element $s \neq 1$ of S . Then $\pi(S) \neq 1$, so $\pi|_S$ is injective because S is simple.

Put $\Gamma = \pi(P)$, $t = \pi(u_1)$, and $J = t^{-1}\pi(S)t$. Since S commutes with $P_1 = u_1 P u_1^{-1}$, the subgroup J centralizes Γ , and $t\Gamma t^{-1} = \pi(P_1) \leq \Gamma$, so $t \in \text{Comp}_{G_0}(\Gamma)$. Moreover $tJt^{-1} = \pi(S) \leq \Gamma$, and Γ has property (T) as a quotient of P .

Lemma 4.3. *Let $\rho: G_0 \rightarrow L$ be a surjective homomorphism with $\rho(\pi(S)) \neq 1$. Then $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$.*

Proof. Write $\Gamma_L = \rho(\Gamma)$, $t_L = \rho(t)$, and $J_L = \rho(J)$. Then $t_L \Gamma_L t_L^{-1} \leq \Gamma_L$, J_L centralizes Γ_L , and $t_L J_L t_L^{-1} = \rho(\pi(S)) \leq \Gamma_L$. For $c \in J_L$ and $\ell \in \Gamma_L$, the commutator $[t_L c t_L^{-1}, \ell]$ lies in $\mathfrak{D}_L(\Gamma_L)$ by (1). As c ranges over J_L , the element $t_L c t_L^{-1}$ ranges over $\rho(\pi(S))$, and ℓ may be taken in $\rho(\pi(S)) \leq \Gamma_L$; hence the commutator subgroup of $\rho(\pi(S))$ lies in $\mathfrak{D}_L(\Gamma_L)$. Since $\rho(\pi(S)) \neq 1$ and S is simple, $\rho(\pi(S)) \cong S$ is perfect, so $\rho(\pi(S)) \leq \mathfrak{D}_L(\Gamma_L)$. $\square$

Proof of Theorem C. Let N be the normal closure of $\pi(S)$ in G_0 . It is nontrivial, so Lemma 4.2 applied to G_0 , N , and $F = \{1, \pi(s)\}$ gives $q: G_0 \rightarrow Q$ with Q two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $q(N) = Q$, and $q(\pi(s)) \neq 1$. The group Q has property (T) as a quotient of G_0 .

Let $r: Q \rightarrow L$ be a surjection onto a nontrivial group and put $\rho = r \circ q$. Since $q(N) = Q$, the normal closure of $\rho(\pi(S))$ in L is $\rho(N) = L$; as $L \neq 1$, this forces $\rho(\pi(S)) \neq 1$. By Lemma 4.3, $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$, and since $\mathfrak{D}_L(\rho(\Gamma))$ is normal in L and contains a normal generating set of L , it equals L . The group L has property (T) as a quotient of Q , and $\rho(\Gamma)$ has property (T) as a quotient of Γ . Theorem A, applied to L with the subgroup $\rho(\Gamma)$ and $K = L$, gives $\text{Rad}_{\text{MF}}(L) = L$. Taking r the identity gives $\text{Rad}_{\text{MF}}(Q) = Q$, and a nontrivial group equal to its own MF radical is not MF. $\square$

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USE OF AI AND FORMAL METHODS

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